

Systems-oriented Software Engineering student specializing in backend systems, Python, and performance optimization. Built production-grade applications using Python, Svelte, and React, with hands-on experience in Linux environments and AI agent workflows. Seeking software engineering roles to apply strong system design and problem-solving skills.

SKILLS

Languages: *Python*, TypeScript, C++, Rust, *Bash*, C#
Frameworks: *SvelteKit*, React, Next.js, Vue, *Django*, Flask, Node, Kotlin, Jetpack Compose, Flutter.
AI/ML: LangGraph, LangChain, PyTorch.
Tools: *Git*, AWS, Google Cloud, Azure, *Linux*, *Docker*, Unity Engine, Godot, Blender.

PROJECTS

- Kepler Zero** ([kepler-zero.onrender.com](#) | [HarshNarayanJha/kepler-zero](#))
Trained a classifier on imbalanced exoplanet data where accuracy fails, optimizing for precision and mitigating skew via transformations; deployed a containerized inference server.
- KabuAI** ([kabuai.onrender.com](#) | [HarshNarayanJha/KabuAI](#))
Built a LangGraph-based AI agent in Python with SSE streaming, improving query relevance 35% via sentiment-aware routing.
- Freyza Employee Systems** ([github.com/FreyzaEmployee](#))
Engineered backend and mobile platform from scratch, streamlining workflows and reducing manual workload by over 60%.
- PYQNow** ([pyqnow.netlify.app](#))
Created a Vue.js/Express portal for 600+ students with responsive UI and automated exam paper scraping.
- Echo - Notepad** ([echo-notepad.netlify.app](#) | [HarshNarayanJha/echo-server](#))
Architected a real-time collaborative system using WebSockets, achieving sub-100ms latency via efficient message handling.

EXPERIENCE

- InPracSys**, Established Healthcare IT — *Fullstack Engineer* — May 2026 - Ongoing
 - Modernized** legacy applications and critical business software in modern software stack improving efficiency and security.
 - Engineered** new features required by the clients hammering down exact business logic and design requirements.
 - Assisted** senior developers and helped migrate legacy deployment flows towards modern CI/CD pipelines and git usage.
- CoRider**, Startup — *Software Developer Engineer* — March 2025 - July 2025
 - Developed** core car-sharing features (DMs, Ask to Verify, and Recurring Trips), enhancing safety and retention of *1k+ users*.
 - Optimized** ride-hosting workflows, reducing listing time by 30% and increasing search-to-match efficiency by 20% through a redesigned *Connections* and date-filtering system.
 - Architected** and deployed scalable backend services using Flask, MongoDB, and MySQL on AWS, reducing API latency by 40% and infrastructure costs by 15%.

OPEN SOURCE

- Refined the Linux desktop experience by submitting patches to *Hyprland*, improving compositor and user experience.
- Implemented XML parsing routine in IA’s *Zeno*, removing external dependency to handle large-scale crawling operations.
- Added a few QoL features to the *Zed Editor* and *Microsoft Terminal*.
- Collaborated with *Unity Technologies* and a community of nearly 100 people to make an open-world RPG game.

EDUCATION

- Birla Institute of Technology**, Jharkhand — *B.Tech* — 2023 - 2027
CGPA: 8.96 — Coursework: Linear algebra, Operating Systems, Computer Architecture, DSA, Database.
- Kendriya Vidyalaya No. 1**, AFS, Darbhanga, Bihar — *10+2* — 2021 - 2023
94.4% in 10th — 90% in 12th—Coursework: Physics, Mathematics, English, Basic Computing, Python, SQL.

ACHIEVEMENTS

- Advent of Code:** Ranked 322 (top ~0.01%) in Advent of Code, consistently solving advanced algorithmic challenges since 2022.
- HarvardX CS50X:** CS50’s Introduction to Computer Science. Awarded by Harvard University (via edX) on completion of CS50.
- University of Helsinki - Fullstackopen:** Completion of 5 ECTS online course with grade 2.
- Honorable mention in SWIFF:** The Honorable mention Letter for the animation short, “The Cycle”, by the SWIFF.